Unit 1 Homework — Real Numbers

Complete independently · Converting · Classification · Squares · Estimating Roots · Ordering

ANSWER KEY

A. Converting Fractions ↔ Decimals

Problem 1

Convert $\frac{1}{4}$ to a decimal.

0.25

Problem 2

Convert $\frac{2}{3}$ to a decimal.

$0.\overline{6}$

Problem 3

Convert 0.8 to a fraction in simplest form.

$\frac{4}{5}$

Problem 4

Convert $\frac{7}{8}$ to a decimal.

0.875

Problem 5

Convert 0.125 to a fraction in simplest form.

$\frac{1}{8}$

Problem 6

Convert $\frac{5}{9}$ to a decimal.

$0.\overline{5}$

B. Classifying Numbers

Problem 1

Is √20 rational or irrational?

irrational

Problem 2

Is √49 rational or irrational?

rational

Problem 3

Is $-\frac{3}{4}$ rational or irrational?

rational

Problem 4

Is 0.1010010001... (non-repeating pattern) rational or irrational?

irrational

Problem 5

Name the smallest set of numbers that −8 belongs to: natural, whole, integer, or rational?

integer

Problem 6

Is √81 rational or irrational?

rational

C. Perfect Squares

Problem 1

What is 7^2?

49

Problem 2

What is 15^2?

225

Problem 3

What is √169?

13

Problem 4

What is √196?

14

D. Estimating Square Roots

Problem 1

Between which two consecutive integers does $\sqrt{40}$ fall?

6 and 7

Problem 2

Estimate $\sqrt{40}$ to the nearest tenth.

6.3

Problem 3

Between which two consecutive integers does $\sqrt{130}$ fall?

11 and 12

E. Ordering Rational & Irrational Numbers

Problem 1

Order from least to greatest: $\frac{7}{2}$, $\sqrt{11}$, 3.4

$\sqrt{11}$, 3.4, $\frac{7}{2}$

Problem 2

Which is greater: $\frac{5}{3}$ or $\sqrt{3}$?

$\sqrt{3}$

Problem 3

Which is greater: $\sqrt{5}$ or 2.2?

$\sqrt{5}$

Problem 4

Order from least to greatest: $-\sqrt{4}$, $-\frac{3}{2}$, -1

$-\sqrt{4}$, $-\frac{3}{2}$, -1